

# Skills, Inequality, and Referrals\*

Experimental Design

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## 1 Background and Setting

*Inequality and Social Class.* Colombia has consistently ranked as one of the most unequal countries in Latin America (Barreto Herrera et al., 2024), with the richest decile earning 50 times more than the poorest decile (ECLAC, 2023). This economic disparity is reflected by a highly stratified society, where social classes remain separated by education, family background, lifestyle, occupation, power, and geographic residence.

*Stratum system.* In 1994, Colombia introduced a nationwide classification system dividing the population into 6 strata based on housing characteristics and neighborhood amenities. This system, initially designed for utility subsidies from higher strata (5 and 6) to support lower strata (1 to 3), now extends to university fees and social program eligibility. While imperfect -because it disregards the income of residents, this system largely aligns with and potentially reinforces existing social class divisions (Guevara S & Shields, 2019; Uribe-Mallarino, 2008). We use this exogenous cutoff as the measure

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\*We obtained Institutional Review Board approvals from NYU Abu Dhabi (HRPP 2024-50) and the University of Luxembourg (ERP 24-028). The study design was preregistered in the OSF Registries prior to data collection (see <https://doi.org/10.17605/OSF.IO/V9T3W>).

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of social class in our experiment: Students in strata 1 to 3 are categorized as low-SES, and those in strata 4 to 6 as high-SES.

*Universidad Autónoma de Bucaramanga.* Our study takes place at UNAB, a medium-sized private university in Bucaramanga, Colombia, with approximately 10,000 enrolled students. UNAB’s student body is notably diverse, with slightly more than half of the students classified as low-SES (see Appendix Figure A.1 for a detailed stratum distribution). This diversity reflects a broader trend in Colombian society, where even lower middle-class families (stratum 2 and 3) prioritize university education for their children despite the financial burden (Hudson & Library of Congress, 2010, p. 103). The university environment provides a unique opportunity for collaborative inter-class contact on equal status, whose positive effects on discrimination are casually documented in numerous studies (Lowe, 2021; Mousa, 2020; Rao, 2019). We draw our sample from a cohort of students taking two semester-long courses. Taking both courses are a requirement to graduate for all students at UNAB, ensuring diversity in academic programs and social classes. However, this approach may introduce some selection bias due to the sorting of students from different social classes into specific study programs.

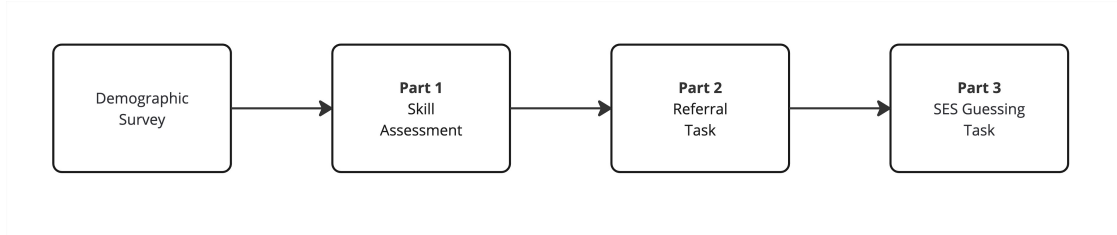
## 2 Design

We designed a referral experiment between students from different social classes, and experimentally varied the incentives to refer a low-SES peer. The study design consists of a single experiment with three sequential parts with sessions organized at the classroom level (see Figure 1). The instructions are provided in Appendix B.

### 2.1 Skill Assessment

To understand the basis for referral decisions, we collect objective measures of cognitive and social skills. These two distinct skills are crucial for the labor market and suitable

Figure 1: Experiment Timeline



*Note:* Participants first complete incentivized skill tests, then refer classmates for skills. In the final part, they guess the social class of their peers. This order is implemented in all sessions.

in a context with university students.<sup>1</sup> This approach ensures that we observe the distribution of skills in our sample, ask for referrals on these skills, and compare how well the same individuals refer peers for a specific skill. Participants perform two incentivized skill tests. They have 5 minutes to complete each test. We provide test-specific instructions and an example item before participants begin. Correctly solved items increase chances to earn a fixed bonus.<sup>2</sup>

*Cognitive Skills.* We use Raven’s Progressive Matrices to measure cognitive skills (Raven, 1936; Raven, Raven, & Court, 1976). Raven’s test is a well-established measure of fluid intelligence, i.e., an individual’s capacity to reason and solve problems in novel situations independent of past knowledge (Schilbach, Schofield, & Mullainathan, 2016). In this test, participants see series of images where there is a pattern with a piece that has been intentionally removed. They are tasked with choosing the piece that completes the pattern among available options. For each image, there is only one correct answer. We implement an 18-item version featuring increasingly difficult questions, with 6 response options for the first 9 items and 8 thereafter.

<sup>1</sup>For the relationship between wages and social skills conditional on cognitive skills, see Deming (2017); Edin, Fredriksson, Nybom, and Öckert (2022); Kuhn and Weinberger (2005); Weinberger (2014).

<sup>2</sup>The tests are presented in a randomized order. No performance feedback is provided. Participants see one item at a time and cannot return to previous screens once they start a test. They are not required to answer items and can skip them if they choose to do so. We elicit beliefs about performance after each test.

*Social Skills.* We measure social skills with the Multiracial Reading the Mind in the Eyes Test (MRMET) from Kim et al. (2022).<sup>3</sup> The test is an established measure for the ability to recognize emotions in others, and it has been previously used in economic experiments (van Leeuwen et al., 2018; Weidmann & Deming, 2021). MRMET consists of photos of human faces portraying different emotions, cropped so that only the eye region is visible. Participants must choose the emotion that best describes the photo from the available answers. For each photo, there is only one correct answer and 4 response options. We administer the first 36 items in MRMET.

## 2.2 Referral Task

We create a referral task to measure bias against low-SES peers. For each skill, participants make incentivized referrals by nominating classmates. We first explain the measured skill accompanied by an example test item. On the following page, we provide an alphabetically ordered list of all classmates. Participants make three referral choices per skill. They are instructed to exclude themselves from referrals. A classmate may be nominated once per triad. The order in which participants refer for a skill test is randomized. We incentivize referrals with classroom-level performance rankings. The three highest-scoring classmates are designated as the top 3 for a skill. Referrers are eligible for a fixed bonus for referrals among the top 3.<sup>4</sup>

*Treatment.* We have one between subject treatment, and it varies the top 3 selection. In the **Baseline** treatment, the top 3 selection is based solely on performance ranking, regardless of other participant characteristics. The **Quota** treatment modifies the top 3 selection to prioritize low-SES individuals. We reserve the first spot in the top 3 for the highest-scoring low-SES classmate, and assign the remaining two places based on performance. This guarantees at least one low-SES participant in the top 3 per skill.

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<sup>3</sup>We choose MRMET because it is a race- and gender-inclusive test suitable for application in non-WEIRD (Western, Educated, Industrial, Rich, Democratic) populations like the one we sample from. The test is based on the original RMET (Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001).

<sup>4</sup>We solve ties among the top 3 randomly. We describe only the top 3 selection mechanism and provide no feedback about the top 3 composition to participants.

Participants are informed about the top 3 selection mechanism before making referral choices (Appendix Figure A.5 provides illustrations explaining the treatments). Assignment to the treatment is at the individual level within each classroom. This allows comparing the effect of the treatment while keeping the referral choice set constant.

### 2.3 SES Guessing Task

Participants make guesses about the anticipated SES of their classmates. We inform participants that a computer algorithm randomly selects three students belonging to strata 1, 2, or 3. They are tasked with nominating the people they believe the computer could choose at random (Appendix Figure A.6 provides the illustration explaining the task). Participants select three classmates from an alphabetically ordered list containing all their classmates. This task measures the ability to distinguish SES independent of test performance, as SES identification is relevant to our study.

## 3 Sample, Incentives, and Procedure

*Sample.* We invited 849 UNAB undergraduate students to participate in the experiment. Our final sample consists of 713 individuals who completed the study protocol, resulting in an 83% participation rate. Table 1 presents key demographic characteristics and academic performance indicators across treatments (Appendix Table A.1 compares the sample and missing students). The sample is well-balanced between the Baseline and Quota conditions. We observe no statistically significant differences in any of the reported variables (all  $p$  values  $> 0.1$ ), indicating balanced randomization. The sample is characterized by a majority of lower-SEB students. About one-third of the sample are first-generation college students. The gender distribution is balanced. The mean GPA of 3.95 is consistent across both treatments.

*Incentives.* Participants could earn bonuses worth 100,000 Pesos (about 26 US Dollars) in each part of the experiment. In the first part, we incentivized performance in the skill tests. 20% of participants were eligible for the bonus. We randomly picked one skill test

for each eligible participant and drew a number between 1 and 100. The participant received the bonus if the percentage of correct answers in the selected test exceeded the drawn number. Chances of earning the bonus increased with each correctly solved question by 5.5% ( $=1/18$ ) for the Cognitive Skill test and by 2.78% ( $=1/36$ ) for the Social Skill test. In the second part, we incentivized referrals among the top 3 performers. 40% of participants were eligible for the bonus. We randomly selected one skill test and one referral for each eligible participant. The participant received the bonus if their referral was among the top 3. In the third part, we incentivized the correct identification of low-SES classmates. 20% of participants in each classroom were eligible for the bonus. We randomly selected one guess for each eligible participant. The participant received the bonus if their guess correctly identified a low-SES classmate.

*Procedure.* Data collection occurred during the last weeks of April 2024. Our local partner, the Social Bee Lab, coordinated scheduled classroom visits and recruited research assistants to administer the experiment. Students present in class on the scheduled visit dates participated. Each classroom visit constituted a separate session. There were in total 35 sessions. Participants accessed the Qualtrics-based experiment using their smartphones during these visits. The median time to complete the survey was 20 minutes, with a compensation of \$26 for 117 lottery winners.

Table 1: Comparison of demographic characteristics and academic outcomes across treatment conditions

|                             | <b>Baseline</b> | <b>Quota</b> | <b><i>p</i></b> |
|-----------------------------|-----------------|--------------|-----------------|
| Low-SES                     | 59%             | 55%          | 0.297           |
| Cognitive skill (out of 18) | 10.04           | 10.27        | 0.322           |
| Social skill (out of 36)    | 18.45           | 18.50        | 0.886           |
| GPA                         | 3.95            | 3.95         | 0.828           |
| Entry Exam                  | 61.85           | 62.17        | 0.638           |
| Age                         | 19.33           | 19.02        | 0.228           |
| Female                      | 52%             | 47%          | 0.195           |
| First Generation            | 34%             | 37%          | 0.386           |
| Ethnic Minority             | 1%              | 3%           | 0.133           |
| Rural Community             | 30%             | 27%          | 0.308           |
| Has Scholarship             | 1%              | 1%           | 0.916           |
| # Semesters at UNAB         | 3.18            | 3.17         | 0.916           |

*Note:* Low SES, female, first generation, ethnic minority, rural, and has scholarship represent binary outcomes and *p*-values for these variables are from two-sample tests of proportions. For continuous variables (age, GPA, cognitive skill, social skill, entry exam, and semester), *p*-values are from two-sample t-tests with equal variances. All reported *p*-values are two-tailed. Sample sizes vary slightly across measures due to missing data.

## 4 Empirical Analysis

### 4.1 Performance

Our first goal is understanding which individuals get referred more often. We consider the fraction of referrals received by total number of referrals that could have been received as our dependent variable (see Appendix Figure A.2 for the distribution). Because every referrer nominates 3 classmates per skill, analyzing only the extensive margin, i.e., whether an individual gets a referral, is not very informative.<sup>5</sup> Our approach combines the intensive and extensive margins and also makes comparisons across classrooms with different sizes easier.<sup>6</sup> Our three dependent variables are fractions of: (i) cognitive skill referrals, (ii) social skill referrals, and (iii) all referrals.

Under performance pay, higher performing classmates in skill tests should collect more referrals. However, it is not clear whether skills can be observed by referrers.<sup>7</sup> We therefore include a common academic performance indicator that may affect our dependent variables, namely the standardized GPA. We control for basic demographics (age, gender, semester) from Table 1 in  $\mathbf{X}_i$  and exclude socioeconomic characteristics which will be examined in the next section. We estimate three referral fractions  $y_i$ :

$$y_i = \beta_0 + \beta_1 GPA_i + \beta_2 Cognitive_i + \beta_3 Social_i + \mathbf{X}_i \boldsymbol{\gamma} + \epsilon_i \quad (1)$$

Table 2 illustrates our first set of findings. Our comparison of interest is between skills and GPA. Conditional on GPA and individual correlates specified above, there is no statistically significant difference between skill level and the fraction of referrals received for that skill (columns 1-2). This holds true also when referrals for both skills are aggregated (column 3). We emphasize aggregated fraction of referrals because most

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<sup>5</sup>Only 86 of the 849 students (10%) never get a referral for either skill.

<sup>6</sup>The number of participating students in a classroom mechanically drives the number of total referrals that could be received by a classmate. By normalizing referrals we focus on differences within instead of between classrooms.

<sup>7</sup>A relevant reason for this could be that we did not provide skill feedback during the experiment. Feedback during sessions could have contaminated referral choices because participants would have incentive to share skill test results.

referrals overlap across skills, as illustrated in Appendix Figure A.3. Point estimates for skills are very close to zero. The 95% confidence intervals rule out that a one standard deviation increase in either skill causes more than a 1 percentage point difference in the fraction of referrals received. This is an order of magnitude smaller margin when compared to the effects of a one standard deviation increase in GPA.

A one standard deviation increase in GPA increases the fraction of referrals by about 3 percentage points on a base rate of 15% across the three dependent variables. This is a modest 20-percent increase in the fraction of referrals received.<sup>8</sup> Smaller point estimates for GPA and lower explained variance suggests our referral model performs better for cognitive skill. Social skill referrals are seemingly made with other considerations not entirely captured in the model. The results suggest participants are not referring classmates who score high on skill tests but refer instead based on their academic performance as measured by GPA.<sup>9</sup> This may be due to classmates being able to observe academic performance of peers but not their skill test results. To expand a bit on the implications of this finding, we show that there is little correlation between skills and GPA (see Appendix Table A.3). A strategy that consists of referring high-GPA students to identify high-skill peers seems to have backfired. This explains the lack of skill differences between the sample and referrals observed in our data.

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<sup>8</sup>The average participant in the average classroom with about 21 students receives 2.3 referrals. A one standard deviation increase in GPA increases this number to 2.8 referrals.

<sup>9</sup>Our findings are robust to restricting the observations to the baseline/quota treatments and including classroom fixed effects (see Appendix Table A.2).

Table 2: Percentage of referrals conditional on GPA and skills

|                     | (1)                  | (2)                  | (3)                  |
|---------------------|----------------------|----------------------|----------------------|
| Dep. Var.           | Cognitive            | Social               | All                  |
| GPA                 | 3.303***<br>(0.463)  | 2.759***<br>(0.452)  | 3.031***<br>(0.419)  |
| Cognitive           | 0.554<br>(0.473)     | 0.000<br>(0.446)     | 0.277<br>(0.430)     |
| Social              | 0.668*<br>(0.403)    | 0.235<br>(0.416)     | 0.451<br>(0.372)     |
| Constant            | 14.977***<br>(2.252) | 14.042***<br>(2.614) | 14.509***<br>(2.238) |
| Individual controls | Yes                  | Yes                  | Yes                  |
| Observations        | 665                  | 665                  | 665                  |
| R-squared           | 0.105                | 0.094                | 0.111                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Dependent variables are the percentage of referrals received relative to all referrals for cognitive (1), social (2), and all referrals (3). Controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data.

## Robustness

In this section we focus on various other specifications of our model in Equation 1. In Table 3, we consider cognitive skill and incrementally add independent variables. A one standard deviation increase in cognitive skill test score increases the fraction of cognitive skill referrals by about 1.1 points on a base rate of 13% (column 1). This effect disappears when GPA is added in the model (column 2) and the 95% confidence intervals rule out that a one standard deviation increase in cognitive skill causes more than a 1 percentage point difference in the fraction of referrals received. Our GPA estimate without controls causes a 3 percentage points increase on a base rate of 12.5%,

consistent with the specification in Table 2. Finally, with the addition of individual controls (age, gender, semester) and social skill in column 3, we obtain the estimates in 2 for cognitive skill. Adding individual controls does not change our performance estimates in a meaningful way.

Table 3: Percentage of cognitive referrals conditional on skills

|                     | (1)                  | (2)                  | (3)                  |
|---------------------|----------------------|----------------------|----------------------|
| Dep. Var.           | Cognitive            | Cognitive            | Cognitive            |
| Cognitive           | 1.111***<br>(0.427)  | 0.677<br>(0.433)     | 0.554<br>(0.473)     |
| GPA                 |                      | 3.187***<br>(0.443)  | 3.303***<br>(0.463)  |
| Social              |                      |                      | 0.668*<br>(0.403)    |
| Constant            | 12.671***<br>(0.404) | 12.483***<br>(0.399) | 14.977***<br>(2.252) |
| Individual controls | No                   | No                   | Yes                  |
| Observations        | 713                  | 668                  | 665                  |
| R-squared           | 0.010                | 0.091                | 0.105                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Dependent variable is the percentage of cognitive referrals received relative to potential referrals. Controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data. All continuous variables are standardized.

In Table 4, we do the same exercise for social skill and incrementally add independent variables. A one standard deviation increase in social skill test score does not increase the fraction of social skill referrals (column 1). This null-effect persists when GPA is added in the model (column 2) and the 95% confidence intervals rule out that a one standard

deviation increase in social skill causes more than a 0.5 percentage point difference in the fraction of referrals received. Our GPA estimate without controls causes a 2.7 percentage points increase on a base rate of 12.6%, consistent with the specification in Table 2. Finally, with the addition of individual controls (age, gender, semester) and cognitive skill in column 3, we obtain the estimates in Table 2 for social skill. Adding individual controls does not change our performance estimates in a meaningful way, but increases the variance explained by the model when compared to cognitive skill. This is mostly due to female participants receiving significantly more referrals for social skill.

Table 4: Percentage of social referrals conditional on skills

|                     | (1)                  | (2)                  | (3)                  |
|---------------------|----------------------|----------------------|----------------------|
| Dep. Var.           | Social               | Social               | Social               |
| Social              | 0.554<br>(0.389)     | 0.284<br>(0.408)     | 0.235<br>(0.416)     |
| GPA                 |                      | 2.773***<br>(0.428)  | 2.759***<br>(0.452)  |
| Cognitive           |                      |                      | 0.000<br>(0.446)     |
| Constant            | 12.785***<br>(0.403) | 12.587***<br>(0.396) | 14.042***<br>(2.614) |
| Individual controls | No                   | No                   | Yes                  |
| Observations        | 713                  | 668                  | 665                  |
| R-squared           | 0.003                | 0.068                | 0.094                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Dependent variable is the percentage of social referrals received relative to potential referrals. Controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data. All continuous variables are standardized.

## 4.2 Treatment Effect

In the following empirical specification, we document whether Baseline and Quota treatments cause referral shares to change, controlling for GPA and classroom SES composition. Our dependent variable is the fraction of referrals received by total number of referrals as specified before. We consider referrals for both skills combined and now decompose it by treatment to compare referral shares depending on the referrer’s treatment. We estimate referral percentage  $y_i$ :

$$y_i = \beta_0 + \beta_1 GPA_i + \beta_2 SES_i + \beta_3 ClassroomSES_i + \mathbf{X}_i\boldsymbol{\gamma} + \epsilon_i \quad (2)$$

In addition to GPA and individual controls (age, gender, semester) in  $\mathbf{X}_i$ , we include referral SES and the percentage of low SES classmates as independent variables as specified in our preregistration.<sup>10</sup>

Table 5 illustrates our findings. Our comparison of interest is estimates across treatments. We obtain the largest estimates for GPA in column (1), documenting the effect of performance-pay incentives in the Baseline treatment without regard for social class. A one standard deviation increase in GPA increases the fraction referrals received from participants in the Baseline condition by 3.8 percentage points on a base rate of 18%, a 21 percent increase. This estimate is slightly smaller in the Quota condition (column 2): A one standard deviation increase in GPA only increases the fraction of referrals by 17 percent.

We find no bias against low SES in either treatment, and also when treatments are combined as a consequence. Point estimates for low SES dummy are close to zero in all three columns. The 95% confidence intervals rule out that low SES causes more than a 1.5 percentage point difference in the fraction of referrals received. Smaller point estimates and substantially lower explained variance in column 2 suggests our model is better at capturing the referral decisions in the Baseline treatment. Further, the Quota

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<sup>10</sup>The relationship between social class and referrals may operate through other channels than socioeconomic status, for example being a first-generation college student, having rural background, etc., which correlate strongly with low SES. We focus on the effect of social class through SES only.

treatment causes no meaningful increase in the referrals received by low SES classmates when controlling for GPA and classroom SES composition, contrary to our hypothesis. In the next section we expand on this finding by decomposing referrals further by referrer SES. This will help us identify whether there is social class bias between peers sharing the same social class.

Table 5: Percentage of referrals across treatment conditions

|                     | (1)                  | (2)                  | (3)                  |
|---------------------|----------------------|----------------------|----------------------|
| Dep. Var.           | Baseline             | Quota                | All                  |
| GPA                 | 3.769***<br>(0.510)  | 2.464***<br>(0.511)  | 3.066***<br>(0.413)  |
| Low SES             | -0.489<br>(0.907)    | -1.079<br>(1.010)    | -0.784<br>(0.764)    |
| % Low SES in class  | -0.037<br>(0.040)    | -0.007<br>(0.041)    | -0.022<br>(0.033)    |
| Constant            | 18.003***<br>(3.802) | 15.191***<br>(3.679) | 16.659***<br>(2.919) |
| Individual controls | Yes                  | Yes                  | Yes                  |
| Observations        | 665                  | 665                  | 665                  |
| R-squared           | 0.119                | 0.044                | 0.110                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage of referrals received relative to potential referrals. Column (1) shows results for baseline treatment, column (2) for quota treatment, and column (3) for all referrals. Controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data.

## Robustness

In this section we focus on various other specifications of our model in Equation 2. In Table 6, we consider referrals in Baseline treatment and incrementally add independent

variables. Low SES does not change the fraction of referrals received in the Baseline treatment (column 1). This finding holds when GPA is added in the model (column 2) and the 95% confidence intervals rule out that low SES causes more than a 1.5 percentage point difference in the fraction of referrals received. In column 3, adding the percentage of low SES peers in the classroom does not change estimates. A one percentage point increase of low SES classmate share does not change the fraction of referrals received nor contribute to the variance explained by the model. Finally, with the addition of individual controls (age, gender, semester) in column 4, we obtain the estimates in 5 for the Baseline condition. Adding individual controls does not change our estimates in a meaningful way.

Table 6: Percentage of baseline referrals with progressive controls

|                     | (1)                  | (2)                  | (3)                  | (4)                  |
|---------------------|----------------------|----------------------|----------------------|----------------------|
| Dep. Var.           | Baseline             | Baseline             | Baseline             | Baseline             |
| Low SES             | -0.586<br>(0.909)    | -0.492<br>(0.906)    | -0.492<br>(0.917)    | -0.489<br>(0.907)    |
| GPA                 |                      | 3.718***<br>(0.492)  | 3.718***<br>(0.496)  | 3.769***<br>(0.510)  |
| % Low SES           |                      |                      | 0.000<br>(0.038)     | -0.037<br>(0.040)    |
| Constant            | 13.394***<br>(0.687) | 13.135***<br>(0.689) | 13.122***<br>(2.272) | 18.003***<br>(3.802) |
| Individual controls | No                   | No                   | No                   | Yes                  |
| Observations        | 713                  | 668                  | 668                  | 665                  |
| R-squared           | 0.001                | 0.094                | 0.094                | 0.119                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage of referrals received relative to potential referrals in the baseline condition. Individual controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data.

In Table 7, we consider referrals in the Quota treatment and incrementally add independent variables. Low SES has a negative but insignificant effect on the fraction of referrals received in the Quota treatment (column 1). This finding persists when GPA is added to the model (column 2), though the magnitude of the SES coefficient decreases slightly. The 95% confidence intervals rule out that low SES causes more than a 1 percentage point difference in the fraction of referrals received. In column 3, adding the percentage of low SES peers in the classroom does not substantially change estimates. A one percentage point increase in low SES classmate share has a negligible effect on the fraction of referrals received and does not contribute to the variance explained by the model. Finally, with the addition of individual controls (age, gender, semester) in column 4, our estimates remain stable. Adding individual controls only marginally increases the variance explained.

Table 7: Percentage of quota referrals with progressive controls

|                     | (1)                  | (2)                  | (3)                  | (4)                  |
|---------------------|----------------------|----------------------|----------------------|----------------------|
| Dep. Var.           | Quota                | Quota                | Quota                | Quota                |
| Low SES             | -1.231<br>(0.970)    | -0.983<br>(1.000)    | -1.022<br>(1.005)    | -1.079<br>(1.010)    |
| GPA                 |                      | 2.378***<br>(0.483)  | 2.391***<br>(0.490)  | 2.464***<br>(0.511)  |
| % Low SES           |                      |                      | 0.014<br>(0.039)     | -0.007<br>(0.041)    |
| Constant            | 13.552***<br>(0.780) | 13.289***<br>(0.815) | 12.486***<br>(2.353) | 15.191***<br>(3.679) |
| Individual controls | No                   | No                   | No                   | Yes                  |
| Observations        | 713                  | 668                  | 668                  | 665                  |
| R-squared           | 0.002                | 0.037                | 0.037                | 0.044                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage of referrals received relative to potential referrals in the quota condition. Individual controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data.

### 4.3 Social Class Bias

In the following empirical specification, we document social class homophily, controlling classroom SES composition.<sup>11</sup> Our dependent variable is the fraction of referrals received by total number of referrals as specified before, but now decomposed by the social class of referrers. We begin with the share of referrals coming from high SES, then present the equivalent analysis for low SES. We estimate referral percentage  $y_i$  the same way in

<sup>11</sup>See [McPherson, Smith-Lovin, and Cook \(2001\)](#) for a discussion on homophily in general, for evidence on social class homophily see [Chetty et al. \(2022a, 2022b\)](#). For experimental evidence on a different type of homophily (gender) in referrals see [Beaman, Keleher, and Magruder \(2018\)](#).

2.

In Table 8, we examine how the share of referrals received differ by referrer’s SES. Column 1 shows referrals from high SES peers: Low SES decreases referrals by 1.89 percentage points, and is marginally significant ( $p < 0.1$ ). The size of the estimate is 75 percent of the effect of a one standard deviation increase GPA which causes a 2.49 percentage point increase in the referral share. On a base rate of 15.8%, low SES classmates are referred 12 percent less by their high SES peers. In contrast, the estimate for the referral share from low SES peers (column 2) does not significantly differ by low SES, suggesting the homophily exists only between high SES referring other high SES peers. Lower explained variance for high SES referrers suggest our model captures the low SES referral decisions better. The overall effect (column 3) shows low SES students receive 0.78 percentage points fewer referrals. Though this difference is not statistically significant, it is driven by high SES referrers. Academic performance, measured by GPA, remains a strong predictor across all specifications, with slightly larger effects for referrals from low SES peers (3.47 versus 2.49 percentage points). Classroom SES composition has no significant effect on the referral share in any specification.

Table 8: Percentage of referrals by referrer’s socioeconomic background

|                     | (1)                  | (2)                  | (3)                  |
|---------------------|----------------------|----------------------|----------------------|
| Dep. Var.           | High SES             | Low SES              | All                  |
| GPA                 | 2.493***<br>(0.555)  | 3.470***<br>(0.506)  | 3.066***<br>(0.413)  |
| Low SES             | -1.895*<br>(0.996)   | 0.245<br>(0.937)     | -0.784<br>(0.764)    |
| % Low SES           | -0.034<br>(0.041)    | -0.032<br>(0.046)    | -0.022<br>(0.033)    |
| Constant            | 15.797***<br>(3.672) | 17.580***<br>(3.853) | 16.659***<br>(2.919) |
| Individual Controls | Yes                  | Yes                  | Yes                  |
| Observations        | 665                  | 665                  | 665                  |
| R-squared           | 0.056                | 0.091                | 0.110                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage of referrals received relative to potential referrals from high SES (1), low SES (2), and all classmates (3). Individual controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data.

## Robustness

In this section we proceed with various specifications of our model in Equation 2, focusing on the SES of the referrer. Table 9 illustrates our main findings. In column (1) we show the raw homophily effect: Referrals from high SES referrers decrease by 1.3 percentage point if recipient is low SES. This is a 21-percent decrease on a base rate of 6.2 percent. In column (2), we add performance, and our homophily estimate decreases by 0.2 percentage points. A one standard deviation increase in either GPA or entry exam score increases high SES referrals as much as being high SES. In column (3), we find that the homophily

is entirely offset by higher performing low SES referrals. Referrals from high SES referrers increase by 1.4 percentage points for a one standard deviation increase in GPA of a low SES participant.<sup>12</sup> In column (4), we add the share of low SES classmates to control for low SES availability in the choice sets of referrers. Referrals from high SES referrers decrease by 0.8 percentage points for low SES participants when availability and performance are controlled for. This effect has the same size (but opposite sign) as a one standard deviation increase in entry exam scores. At the same time, referrals increase by 1 percentage point for a one standard deviation increase in GPA of a low SES participant. These results suggest that while social class homophily exists, it primarily affects low-performing, low SES students. High-performing low SES students not only overcome this disadvantage but actually receive more referrals from high SES peers than their high SES counterparts with similar performance. Finally, a one percent increase in low SES availability in a classroom with no low SES students (when availability is at 0) mechanically decreases high SES referrals by 1.5 percentage points.

We now compare these results to low SES referrals in Table 10. In column (1), we observe that raw homophily estimate is indistinguishable from zero: The effect does not exist for low SES referrers. Adding performance in column (2), we still find a homophily estimate close to zero, even though the estimates for GPA and entry exam score are comparable for low SES referrals. With the interaction effect in column (3), we observe that the performance effect is uniform, with one standard deviation increase in GPA resulting in between 2.2 to 1.7 percentage points more referrals on a base rate of 6.7 percent. When controlling for the share of low SES classmates in column (4), the results paint a clear picture: Low SES referrers show no homophily in their choices and respond twice as strongly to GPA compared to high SES referrers, while maintaining similar sensitivity to entry exam scores. The classroom composition effect mirrors that of high SES referrers, with a one percent increase in low SES availability in a classroom with only high SES students mechanically increasing low SES referrals by 1.3 percentage points.

Taken together, these results reveal an asymmetric homophily pattern: While high

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<sup>12</sup>This relationship does not hold for entry exam scores and we do not report them for conciseness.

SES referrers show significant SES homophily that can be overcome by academic performance, low SES referrers appear to base their decisions primarily on performance regardless of social class. This suggests that the barriers to cross-class referrals originate from high-SES students, but are overcome through academic achievement in our sample.

Table 9: Percentage of referrals from high SES referrers

|                      | (1)                  | (2)                 | (3)                 | (4)                  |
|----------------------|----------------------|---------------------|---------------------|----------------------|
| Dep. Var.            | High SES             | High SES            | High SES            | High SES             |
| Low SES              | -1.318***<br>(0.465) | -1.112**<br>(0.477) | -1.167**<br>(0.471) | -0.780*<br>(0.434)   |
| GPA                  |                      | 1.035***<br>(0.256) | 0.253<br>(0.411)    | 0.351<br>(0.365)     |
| Entry Exam           |                      | 0.894***<br>(0.277) | 0.948***<br>(0.275) | 0.749***<br>(0.254)  |
| Low SES $\times$ GPA |                      |                     | 1.404***<br>(0.499) | 1.042**<br>(0.450)   |
| Classroom % Low SES  |                      |                     |                     | -0.149***<br>(0.020) |
| Constant             | 6.180***<br>(1.193)  | 5.283**<br>(2.180)  | 5.017**<br>(2.185)  | 14.570***<br>(2.520) |
| Baseline controls    | Yes                  | Yes                 | Yes                 | Yes                  |
| Observations         | 666                  | 633                 | 633                 | 633                  |
| R-squared            | 0.015                | 0.076               | 0.090               | 0.200                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage of referrals received from high SES classmates. Controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data.

Table 10: Percentage of referrals from low SES referrers

|                      | (1)                 | (2)                 | (3)                 | (4)                 |
|----------------------|---------------------|---------------------|---------------------|---------------------|
| Dep. Var.            | Low SES             | Low SES             | Low SES             | Low SES             |
| Low SES              | 0.375<br>(0.554)    | 0.663<br>(0.537)    | 0.681<br>(0.530)    | 0.335<br>(0.519)    |
| GPA                  |                     | 1.931***<br>(0.319) | 2.191***<br>(0.500) | 2.104***<br>(0.482) |
| Entry Exam           |                     | 0.757**<br>(0.318)  | 0.739**<br>(0.320)  | 0.917***<br>(0.313) |
| Low SES $\times$ GPA |                     |                     | -0.466<br>(0.594)   | -0.143<br>(0.570)   |
| Classroom % Low SES  |                     |                     |                     | 0.133***<br>(0.022) |
| Constant             | 7.552***<br>(1.607) | 6.660**<br>(2.697)  | 6.748**<br>(2.708)  | -1.788<br>(2.856)   |
| Baseline controls    | Yes                 | Yes                 | Yes                 | Yes                 |
| Observations         | 666                 | 633                 | 633                 | 633                 |
| R-squared            | 0.029               | 0.111               | 0.112               | 0.172               |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage of referrals received from low-SES classmates. Controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data.

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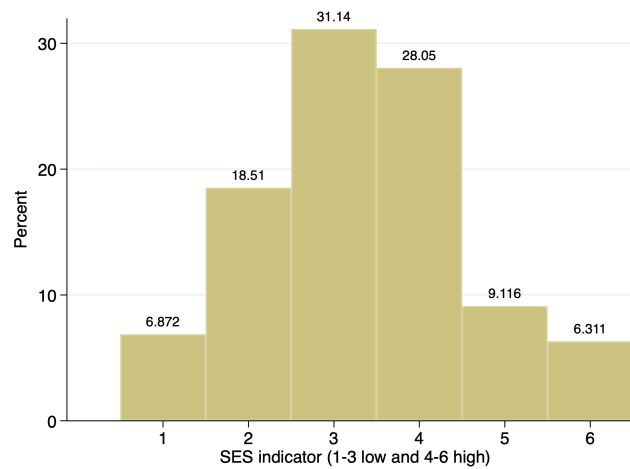
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## A Additional Figures and Tables

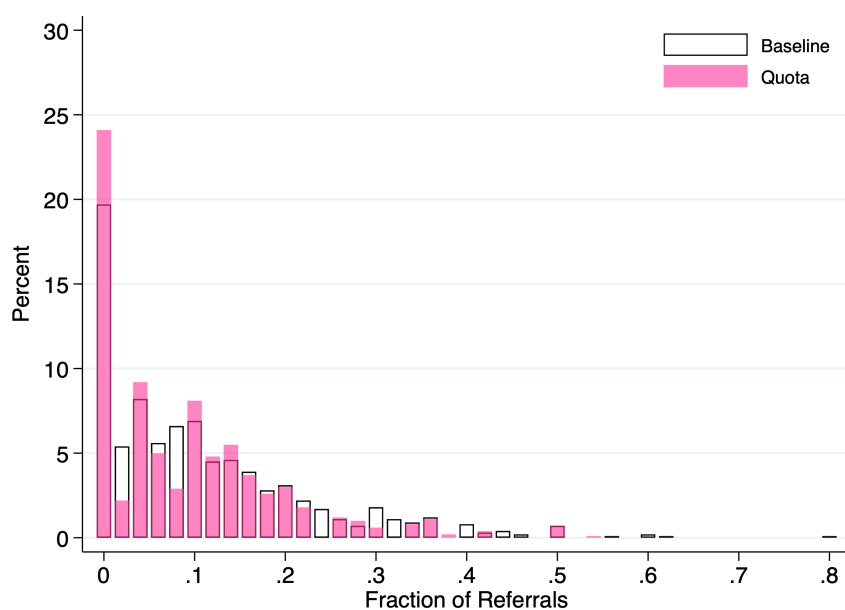
### A.1 Additional Figures

Figure A.1: Stratum distribution of the sample



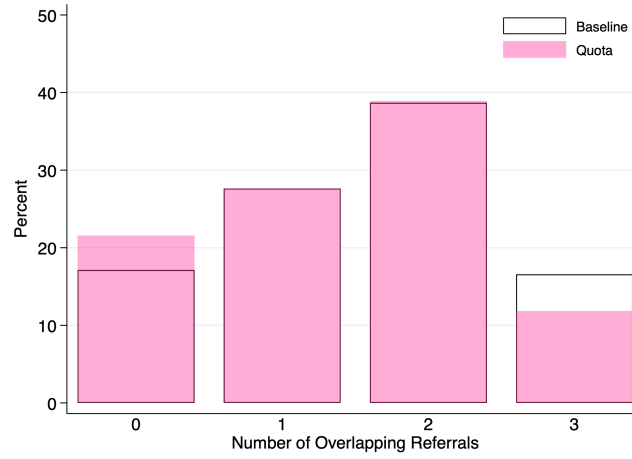
### A.2 Additional Tables

Figure A.2: Fraction of referrals received



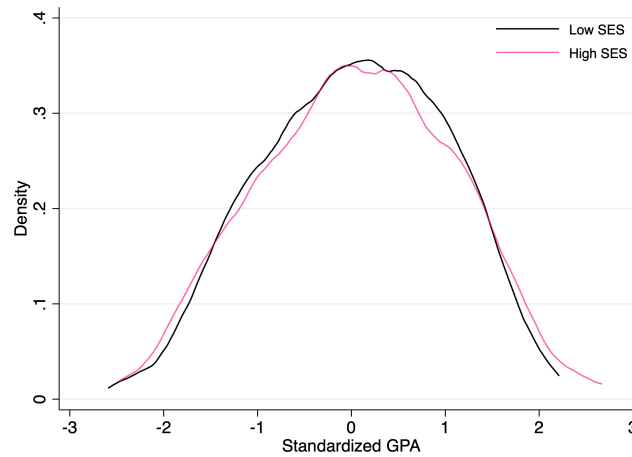
*Note:* The figure shows the fraction of referrals received across cognitive and social skills, disaggregated by baseline and quota treatments. The fraction is calculated as the number of actual referrals received divided by the maximum possible referrals a student could have received. To illustrate: Consider a classroom with 25 students where 20 participated in the experiment, with 10 students in each treatment group. Each student could receive a maximum of 10 referrals (excluding self-referrals) from others in their treatment group. We normalize the total referrals received by this maximum possible number to obtain the fraction shown in the figure. The distribution of referrals is right-skewed with extreme outliers. While the median student across treatments receives 9 percent of possible referrals, those above the 90th percentile receive between 27 and 100 percent.

Figure A.3: Distribution of overlapping referrals



*Note:* The figure shows the distribution of overlapping referrals across conditions, excluding self-referrals. The first bar (value of 0) indicates the percentage of individuals with 6 unique referrals for both skills. The last bar (value of 3) indicates the percentage of individuals with 3 identical referral choices. Most participants have 2 or more overlapping referral choices.

Figure A.4: Distribution of standardized GPA across SES



*Note:* There are no significant differences in standardized GPA between high-SES ( $M = 0.058$ ,  $SD = 1.032$ ,  $n = 286$ ) and low-SES students ( $M = 0.033$ ,  $SD = 0.970$ ,  $n = 382$ ),  $t(666) = 0.32$ ,  $p = 0.746$ ,  $d = 0.025$ .

Table A.1: Comparison of Variables Between the Sample and Missing Students

|                               | Sample | Missing | <i>p</i> |
|-------------------------------|--------|---------|----------|
| Ratio referrals (both skills) | 0.127  | 0.043   | 0.000    |
| GPA (standardized)            | 0.044  | -0.273  | 0.001    |
| Entry Exam (standardized)     | 0.028  | -0.168  | 0.046    |
| # Semesters at UNAB           | 3.171  | 3.188   | 0.884    |
| Age                           | 19.182 | 20.287  | 0.001    |
| Female                        | 49.8%  | 48.5%   | 0.788    |
| Ethnic Minority               | 2.1%   | 4.4%    | 0.114    |
| Rural Community               | 28.8%  | 31.6%   | 0.501    |
| Has Scholarship               | 0.8%   | 0.7%    | 0.899    |

*Note:* Values for female, ethnic minority, rural community, and scholarship represent percentages. All other variables represent means. *p*-values for gender, ethnic, rural, and scholarship are from two-sample tests of proportions. For all other variables, *p*-values are from two-sample t-tests with equal variances. All tests compare the sample and missing students. All reported *p*-values are two-tailed.

Table A.2: Percentage of referrals across specifications conditional on performance

|                     | (1)                  | (2)                  | (3)                  | (4)                  |
|---------------------|----------------------|----------------------|----------------------|----------------------|
| Dep. Var.           | All                  | Baseline             | Quota                | All                  |
| GPA                 | 3.031***<br>(0.418)  | 3.765***<br>(0.515)  | 2.402***<br>(0.513)  | 3.306***<br>(0.529)  |
| Cognitive           | 0.243<br>(0.437)     | 0.278<br>(0.491)     | 0.277<br>(0.545)     | 0.327<br>(0.455)     |
| Social              | 0.412<br>(0.370)     | 0.114<br>(0.426)     | 0.577<br>(0.511)     | 0.159<br>(0.302)     |
| Constant            | 14.873***<br>(2.358) | 15.598***<br>(3.091) | 14.042***<br>(2.952) | 14.421***<br>(2.324) |
| Classroom FE        | No                   | No                   | No                   | Yes                  |
| Individual Controls | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations        | 665                  | 665                  | 665                  | 665                  |
| R-squared           | 0.112                | 0.118                | 0.047                | 0.115                |

*Note:* Robust standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage of referrals received relative to potential referrals. Column (1) shows results for all referrals, column (2) for baseline condition only, column (3) for quota treatment only, and column (4) includes classroom fixed effects. Controls include gender, age and semester. Sample is restricted to participants with complete administrative and experimental data. All continuous variables are standardized.

Table A.3: Correlation between GPA and skills

|           | GPA    | Cognitive | Social |
|-----------|--------|-----------|--------|
| GPA       | 1.000  |           |        |
| Cognitive | 0.085  | 1.000     |        |
| Social    | 0.093* | 0.267*    | 1.000  |

*Note:* Pairwise correlation between GPA and skills. \* indicates significance at the 5% level with Bonferroni correction. Sample is restricted to participants with complete administrative and experimental data.

## B Experiment

*We include the English version of the instructions used in Qualtrics. Participants saw the Spanish version. Horizontal lines indicate page breaks, and clarifying comments are inside brackets.*

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Please enter the password:

[classroom-specific password sent to each participant the day before data collection]

---

### Welcome

Welcome to this study organized by the Social Bee Lab. You have been invited to participate in a survey where you can make a series of decisions. The study takes approximately 20 minutes to complete. During the study, you should not communicate with any other students. If you have any questions at any time, please raise your hand. One of the assistants will help you privately.

In this study, you can win bonus money depending on your choices. In total, we will

draw [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos among the participants of this classroom. It is also possible for the same person to win more than one voucher. The following screens will detail how the bonus draw will be conducted. The UNAB finance office will make the payment of the vouchers through Nequi.

All your decisions in this survey will be anonymized. Therefore, the answers you provide will not affect your grades in this class or your records at the university. We will use your personal information to determine the bonus allocation, but after that, we will remove any data that identifies you.

This survey has several parts. Each of these parts has specific instructions. Please read the instructions for each part carefully because they describe how you can earn bonuses. This study has been approved by the [omitted for anonymous review] on the condition that all the information we provide is true and all the bonuses we offer are real.

On the next screen, we present you with an informed consent form that you must accept to participate in this study.

---

### **Informed Consent**

You have been invited to participate in a study to learn more about how people make decisions in common scenarios.

This study is conducted by [omitted for anonymous review] and the Social Bee Lab at UNAB. The purpose of this study is to broaden our understanding of how people make decisions.

Participation in this study is voluntary. You may opt-out at any time. No known

risks are associated with your participation in this project beyond those of everyday life. Apart from the monetary bonuses that will be drawn, participation has no direct benefits.

The Social Bee Lab is in charge of data collection. Your answers in this study are anonymous and will not be shared with anyone. In addition to your answers, UNAB will provide the Social Bee Lab with administrative records of your courses and your university entrance exam score. Your records, decisions, and your identity will be kept strictly confidential. Data about you collected within the scope of the study are used for scientific purposes only and are treated as strictly confidential. The Social Bee Lab will anonymize your data, and the researcher will analyze it without knowing your identity. All data generated will be stored on the researcher's computer. You have the right to access your personal data and request its deletion. You can exercise this right by contacting the researcher.

If something is unclear or you have any questions, you can contact [omitted for anonymous review].

If you have questions about your rights as a participant, you can contact [omitted for anonymous review].

By continuing to the next screen, you agree to participate in this study.

---

Before you start, please answer these four questions.

What is your gender?

[Male, Female]

What is the socio-economic stratum to which your family belongs?

[Stratum 1 to Stratum 6]

What is your father's highest acquired level of education?

[Primary school, High school, Technical school, Undergraduate, Graduate, Postgraduate, Not applicable.]

What is your mother's highest acquired level of education?

[Primary school, High school, Technical school, Undergraduate, Graduate, Postgraduate, Not applicable.]

---

## **Part 1**

You will now participate in two quizzes, each lasting five minutes. Please try to answer them to the best of your ability.

We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 100.000 pesos in this first part. The steps to allocate the bonuses for Part 1 are explained below.

---

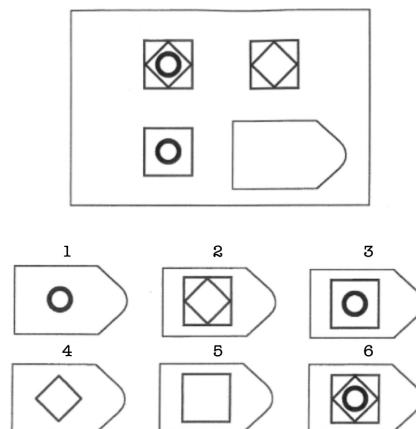
[classroom-specific illustrations explaining the incentive structure]

---

[random assignment to either cognitive or social skills test]

### Test - Cognitive Skill

In this test, you will see a series of images. Below is an example of the images you will solve. At the top of each image, there is a pattern with a piece that has been removed. Your task is to choose which of the six pieces completes the pattern correctly. For each image, there is only one correct piece. Look at the following example:



First, notice a square in the upper left, the upper right, and the lower left. Also, notice that the circle is eliminated when one moves from the upper left to the upper right. Finally, the rhombus is eliminated when moving from the upper left to the lower left. Therefore, the correct piece should eliminate the circle and the rhombus, leaving only a square. So, the correct answer is piece 5.

To give your answer to each image, you must choose the correct option and then continue

to the next screen. After giving your answer you cannot go back.

You will have 5 minutes to complete the test, which consists of 18 images to solve. The percentage of correct answers will determine your chances of winning one of the 100.000 pesos bonuses if you are chosen for the drawing.

---

Are you ready?

Your 5 minutes will start as soon as you move to the next screen.

---

### **Problem 1**

[screenshot of Raven's matrix]

[After participants submit an answer, a new matrix appears on the screen. The sequence of matrices is the same for all participants. Participants cannot return to a previous screen. Participants do not have to provide answers for all 18 matrices.]

---

You have finished the test. You can proceed to the next screen.

---

**How did you do on the test?**

If we randomly choose 10 participants from this classroom, how many people do you think solved fewer correct problems than you?

[Slider from 0 to 10]

---

### **Test - Emotions**

In this test, you will see a series of photographs. Below is an example of the pictures you will see. In each picture, you will see the eyes of a person. Below the picture, you will see four possible emotions that this person is feeling. Your task is to choose which of the four emotions correctly describes what the person is feeling. For each picture, there is only one emotion. Look at the following example:



[Happy, Disappointed, Shocked, Worried]

In this case, the correct answer is: Shocked.

To give your answer to each picture, you must choose the correct option and then continue to the next screen. After giving your answer you will not be able to go back.

You will have 5 minutes to complete the test, which consists of 36 photographs to solve. The percentage of correct answers you get will determine your chances of winning one

of the 100.000 pesos bonuses if you are chosen for the drawing.

---

Are you ready?

Your 5 minutes will start as soon as you move to the next screen.

---

**Photograph 1: Choose the word that best describes the photograph**

[photo from Multiracial Reading the Mind in the Eyes Test]

[After participants submit an answer, a new photo appears on the screen. The sequence of photos is the same for all participants. Participants cannot return to a previous screen. Participants do not have to provide answers for all 36 photos.]

---

You have finished the test. You can proceed to the next screen.

---

**How did you do on the test?**

If we randomly choose 10 participants from this classroom, how many people do you think solved fewer correct photographs than you?

[Slider from 0 to 10]

---

## **Part 2**

At the beginning of this study, all participants took two tests, one on cognitive ability and one on emotions. In this part, we will ask you to recommend the people who in your opinion will score the best on each test.

You may recommend 3 people per test, but you may not recommend yourself.

We will allocate up to [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos for Part 2. The steps for allocating bonuses are explained below.

---

[random assignment to either quota or baseline condition]

[classroom-specific illustrations explaining the incentive structure depending on assignment to either baseline or quota conditions]

---

[random assignment to either cognitive or social skills referral task]

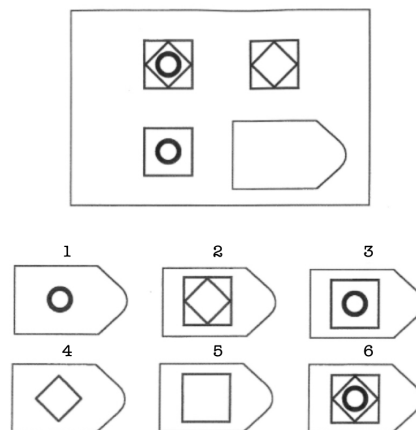
## **Recommendation - Cognitive Skill**

All participants took a test to identify the missing pattern in each image, as in the ex-

Figure A.5: Illustrations for the two conditions



ample below. This test is used to measure general intelligence.



Next, we will present you with a list of the names of all the students in this room. We will ask you to recommend the three people you think will score the highest on the general intelligence test.

If you are chosen by the computer, each of your recommendations in the top 3 increases your chances of winning one of the 100.000 pesos bonuses.

---

Select the students in this classroom who you consider to have the highest scores on the general intelligence test. (Select 3 students)

[Classroom-specific list of all classmate names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

---

### **Recommendation - Emotions**

All participants took a test where they had to identify the emotion that best described the expression of each image as in the example below. This test is used to measure social skills.



Next, we will present you with a list of the names of all the students in this room. We will ask you to recommend 3 people you think will score the highest on the social skills test.

If you are chosen by the computer, each of your recommendations in the top 3 increases your chances of winning one of the 100.000 pesos bonuses.

---

Select the students in this classroom who you consider to have the highest scores on the social skills test. (Select 3 students)

[Classroom-specific list of all the names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

---

### Part 3: Recommendation - Random draw

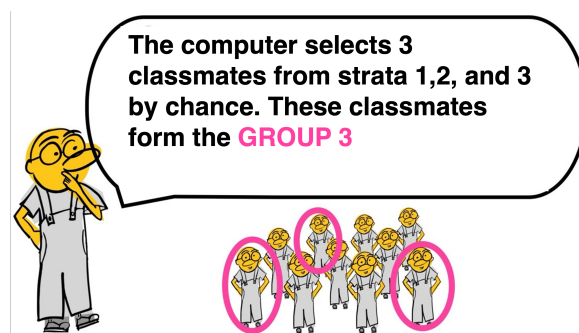
In this part, the computer will randomly choose three students who belong to strata 1, 2, or 3. We will ask you to nominate three people you think the computer will choose.

We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 100.000 pesos for Part 3. The steps for allocating the bonuses are explained below.

---

[classroom-specific illustrations explaining the incentive structure]

Figure A.6: Illustration for the Guessing Task



Select the students in this classroom who belong to strata 1, 2, or 3, who you think will be randomly selected by the computer (Select 3 students).

[Classroom-specific list of all the names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

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#### **Part 4**

Do you want to know your scores on the general intelligence test and the social skills test? We can analyze the data and give you a report that explains your strengths in these two areas. Also, what do these strengths mean, and how can you leverage them for your personal and professional development?

If you want to receive your skills report, we need to contact you again. We also want to be able to invite you to new studies where you can participate for more bonus money. Please indicate if you agree to be contacted again.

[I can be contacted for new studies and to send me my report. I can be contacted to send my report, but not for new studies. No, I do not want to be contacted again.]

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[if participant gives consent to be contacted again]

Please enter your contact email:

[student email]